



VECTOR SPACES AND SUBSPACES

1.1. INTRODUCTION

Students are already familiar with algebraic structures such as groups, rings and fields that involve only internal binary operations. In this chapter we shall study a new algebraic structure called *Vector Space* that involves internal as well as external binary operation, which are defined below. We shall also study subspace of vector space.

Internal and External Binary Operation

Let V and F be two non-empty sets.

1. If in V , $u * v \in V$ for all $u, v \in V$ and $u * v$ is unique, then the binary operation $*$ is an **internal binary operation** on V .

2. If for each $u \in V$ and $a \in F$, $a * u \in V$ and $a * u$ is unique, then $*$ is called an **external binary operation** on V over F .

1.2. VECTOR SPACE

Definition 1. Let F be a field. A **vector space** over F , is a non-empty set V together with two operations (called addition and scalar multiplication) such that for each $u, v \in V$ there is a unique element $u + v \in V$ and for each $a \in F$ and $u \in V$ there is a unique element $au \in V$, and it satisfies the following conditions:

- I. (i) $u + (v + w) = (u + v) + w$, for all $u, v, w \in V$ (Associativity)
- (ii) $u + v = v + u$, for all $u, v \in V$ (Commutativity)
- (iii) There exists an element $0 \in V$ such that $u + 0 = u = 0 + u$, for all $u \in V$ (Existence of identity element)
- (iv) For each $u \in V$, there exists a unique element $-u \in V$ such that

$$u + (-u) = 0 = (-u) + u \text{ (Existence of additive inverse)}$$

II. For any $u, v \in V, a, b \in F$, the following conditions are satisfied :

(i) $a(u + v) = au + av$, for all $a \in F$ and $u, v \in V$.

[Scalar multiplication is distributive over addition in V]

(ii) $(a + b)u = au + bu$, for all $a, b \in F$ and $u \in V$.

[Scalar multiplication is distributive over addition in F]

(iii) $(ab)u = a(bu)$, for all $a, b \in F$ and $u \in V$.

(iv) $1u = u$, for all $u \in V$, where 1 is the multiplicative identity of F .

To specify a vector space, we must be given

- (1) A non-empty set V
- (2) Two operations called *addition* and *scalar multiplication* satisfying the properties of above definition.

The first thing to check is whether (1) and (2) hold or not. If either of this fails, we do not have a vector space.

The advantage of above definition is that it is not concerned with the question of what a vector is. For example, is a vector in $\mathbb{R}^2$ a point, a directed line segment, or a 2×1 matrix. Definition 1 is only concerned with the algebraic behaviour of elements in a vector space. Thus *a vector is now simply an element of a vector space.*

Remarks.

- (i) The elements of F are called **scalars** and the elements of V are called **vectors**. The reader should not confuse the use of the word "vector" with the physical entity having magnitude and direction.
- (ii) The word "vector" is being used to describe any element of the vector space.
- (iii) A vector space V over a field F is denoted by $V(F)$. Instead of saying " V is a vector space over a field F ", we say $V(F)$ or simply V is a vector space.
- (iv) The symbol 0 has been used to denote the zero element of F and the symbol $\mathbf{0}$ has been used to denote the zero element of V .
- (v) The symbol $+$ denotes the addition in V as well as addition in F .
- (vi) The conditions (i) to (iv) in the definition 1 of the vector space show that $\langle V, + \rangle$ is an **abelian group**. Thus we can also define the vector space as follows:

Definition 2. Let $\langle V, + \rangle$ be an **abelian group** and $\langle F, +, \cdot \rangle$ be a field. Define scalar multiplication such that for all $a \in F, u \in V, au \in V$. Then we say that V is a vector space over F if the following conditions are satisfied :

- (i) $a(u + v) = au + av$, for all $a \in F$ and $u, v \in V$
- (ii) $(a + b)u = au + bu$, for all $a, b \in F$ and $u \in V$.
- (iii) $a(bu) = (ab)u$, for all $a, b \in F$ and $u \in V$
- (iv) $1u = u$, for all $u \in V$, 1 being multiplicative identity of F .

1.3. VECTORS IN $\mathbf{R}^n$

Let $\mathbf{R}$ be the set of real numbers. $\mathbf{R}$ can be regarded geometrically as the Euclidean line, i.e. as *Euclidean 1-space*. We know that each ordered pair (a, b) of real numbers is represented by a point in the plane. The set of all such ordered pairs of real numbers is called *Euclidean 2-space* and is denoted by $\mathbf{R}^2$.

Similarly, the set of all ordered triples (a, b, c) of real numbers is called *Euclidean 3-space* and is denoted by $\mathbf{R}^3$ and the set of all n -tuples of real numbers, denoted by $\mathbf{R}^n$, is called *Euclidean n -space*. We write an n -tuple as a $1 \times n$ row matrix. The number of elements (columns) n is the dimension of the row matrix.

For example, the ordered pair $(2, -3)$ belongs to $\mathbf{R}^2$; it is a 2-tuple of dimension two.

The ordered triple $(7, 3, 6)$ belongs to $\mathbf{R}^3$; it is 3-tuple of dimension three.

Definition. An ordered n -tuple of real numbers is called a *real n -vector*.

An ordered array of real numbers can also be written as a column vector. Such a vector is called *column vector*.

For example, $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 7 \\ 3 \\ 6 \end{bmatrix}$ are column vectors and can be written as $(2, -3)$ and $(7, 3, 6)$ respectively.

Note.

- (i) A **plane vector** is an ordered pair (a_1, a_2) of real numbers. The set of plane vectors (all ordered pairs of real numbers) is denoted by V_2 which is a cartesian product $\mathbf{R} \times \mathbf{R}$ and we denote it as $\mathbf{R}^2$.

We say that two plane vectors (a_1, a_2) and (b_1, b_2) are equal iff $a_1 = b_1$ and $a_2 = b_2$. Addition of vectors in $\mathbf{R}^2$ is defined by $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$, where $(a_1, a_2), (b_1, b_2) \in \mathbf{R}^2$

- (ii) A **space vector** is an ordered triple (a_1, a_2, a_3) of real numbers. The set of all plane vectors (all ordered triples of real numbers) is denoted by V_3 which is a Cartesian product $\mathbf{R} \times \mathbf{R} \times \mathbf{R}$ and we denote it as $\mathbf{R}^3$.

(iii) A vector in $\mathbf{R}^n$ (also denoted by $\mathbf{R}_n$) is an ordered n -tuple $(a_1, a_2, \dots, a_n)$ where $a_1, a_2, \dots, a_n$ are real numbers. Thus a n -tuple vector $X = (a_1, a_2, \dots, a_n)$ is called an n -vector and a_i is called the i th component or co-ordinate or entry of X .

Two vectors $X = (a_1, a_2, \dots, a_n)$ and $Y = (b_1, b_2, \dots, b_n)$ are said to be **equal** iff their corresponding components are equal. Thus $X = Y$ iff $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$.

Given two vectors X and Y in $\mathbf{R}^n$, then their **sum** is the vector

$X + Y = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$ and this addition is called vector addition.

If c is a real number (scalar), then the **scalar multiple** $cX = (ca_1, ca_2, \dots, ca_n)$. This multiplication is called **scalar multiplication**.

SOLVED EXAMPLES

Example 1.

Prove that the set of all vectors in a plane over the field of real numbers is a vector space with respect to vector addition and scalar multiplication.

Solution. Let V denotes the set of all coplanar vectors and $\mathbf{R}$ be the field of real numbers.

$\therefore$ The elements of V are the ordered pairs (x, y) , where $x, y \in \mathbf{R}$

i.e., $V = \{(x, y) : x, y \in \mathbf{R}\}$

Internal Binary Operation on V :

For $u, v \in V$ and $u = (u_1, u_2), v = (v_1, v_2)$

$$u + v = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) \in V \quad [\because u_1, u_2, v_1, v_2 \in \mathbf{R}]$$

Thus addition operation is well defined in V . Hence, it is an internal binary operation on V over $\mathbf{R}$.

External Binary Operation on V :

For $u = (u_1, u_2) \in V$ and $u_1, u_2, \alpha \in \mathbf{R}$

$$\alpha(u_1, u_2) = (\alpha u_1, \alpha u_2) \in V$$

Thus the scalar multiplication defined in V is an external binary operation on V over $\mathbf{R}$.

Now, let us prove all the axioms of vector space.

(I) $(V, +)$ is an abelian group.

(i) **Associativity:**

We know that for all $u, v, w \in V$

$$(u + v) + w = u + (v + w)$$

(ii) **Commutativity:**

We know that for all $u, v \in V$

$$u + v = v + u$$

(iii) A vector in $\mathbf{R}^n$ (also denoted by $\mathbf{R}_n$) is an ordered n -tuple $(a_1, a_2, \dots, a_n)$ where $a_1, a_2, \dots, a_n$ are real numbers. Thus a n -tuple vector $X = (a_1, a_2, \dots, a_n)$ is called an n -vector and a_i is called the i th component or co-ordinate or entry of X .

Two vectors $X = (a_1, a_2, \dots, a_n)$ and $Y = (b_1, b_2, \dots, b_n)$ are said to be **equal** iff their corresponding components are equal. Thus $X = Y$ iff $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$.

Given two vectors X and Y in $\mathbf{R}^n$, then their **sum** is the vector

$X + Y = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$ and this addition is called vector addition.

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SOLVED EXAMPLES

Example 1.

Prove that the set of all vectors in a plane over the field of real numbers is a vector space with respect to vector addition and scalar multiplication.

Solution. Let V denotes the set of all coplanar vectors and $\mathbf{R}$ be the field of real numbers.

$\therefore$ The elements of V are the ordered pairs (x, y) , where $x, y \in \mathbf{R}$

i.e., $V = \{(x, y) : x, y \in \mathbf{R}\}$

Internal Binary Operation on V :

For $u, v \in V$ and $u = (u_1, u_2), v = (v_1, v_2)$

$$u + v = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) \in V \quad [\because u_1, u_2, v_1, v_2 \in \mathbf{R}]$$

Thus addition operation is well defined in V . Hence, it is an internal binary operation on V over $\mathbf{R}$.

External Binary Operation on V :

For $u = (u_1, u_2) \in V$ and $\alpha \in \mathbf{R}$

$$\alpha(u_1, u_2) = (\alpha u_1, \alpha u_2) \in V$$

Thus the scalar multiplication defined in V is an external binary operation on V over $\mathbf{R}$.

Now, let us prove all the axioms of vector space.

(I) $(V, +)$ is an abelian group.

(i) **Associativity:**

We know that for all $u, v, w \in V$

$$(u + v) + w = u + (v + w)$$

(ii) **Commutativity:**

We know that for all $u, v \in V$

$$u + v = v + u$$

(iii) **Existence of additive identity :**

For every vector $u \in V$ there exists a zero vector $\mathbf{0} \in V$ such that

$$u + \mathbf{0} = \mathbf{0} + u = u.$$

(iv) **Existence of additive inverse :**

For every vector $u \in V$ there exists a vector $-u \in V$ such that

$$u + (-u) = (-u) + u = \mathbf{0}$$

Thus V is an abelian group with respect to vector addition.

(II) Properties of scalar multiplication in V .

(i) For $u, v \in V$ and $a \in \mathbf{R}$, we have

$$a(u + v) = au + av$$

(ii) For $u \in V$ and $a, b \in \mathbf{R}$, we have

$$(a + b)u = au + bu$$

(iii) For $u \in V$ and $a, b \in \mathbf{R}$, we have

$$a(bu) = (ab)u$$

(iv) For $u \in V$ and $1 \in \mathbf{R}$, we have

$$1u = u.$$

Thus set V of coplanar vectors satisfies all the properties of vector addition and scalar multiplication and hence V is a vector space.

Example 2.

Prove that the set C of all complex numbers (i.e., the set of all ordered pairs of real numbers) is a vector space over the field $\mathbf{R}$ of all real numbers where vector addition is defined by $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$, for all $(x_1, x_2), (y_1, y_2) \in C$ and scalar multiplication is defined by $a(x_1, x_2) = (ax_1, ax_2)$, for all $a \in \mathbf{R}$.

Solution. We observe that C is closed under vector addition and under scalar multiplication.

Now, let us prove all the axioms of vector space.

I. $(C, +)$ is an abelian group**(i) Associativity :**

For all $(x_1, x_2), (y_1, y_2), (z_1, z_2) \in C$, we have

$$\begin{aligned} (x_1, x_2) + [(y_1, y_2) + (z_1, z_2)] &= (x_1, x_2) + (y_1 + z_1, y_2 + z_2) \\ &= (x_1 + y_1 + z_1, x_2 + y_2 + z_2) \\ &= (x_1 + y_1, x_2 + y_2) + (z_1, z_2) \\ &= [(x_1, x_2) + (y_1, y_2)] + (z_1, z_2) \end{aligned}$$

(ii) Commutativity:

For all $(x_1, x_2), (y_1, y_2) \in \mathbf{C}$, we have

$$\begin{aligned}(x_1, x_2) + (y_1, y_2) &= (x_1 + y_1, x_2 + y_2) \\ &= (y_1 + x_1, y_2 + x_2) \\ &= (y_1, y_2) + (x_1, x_2)\end{aligned}$$

(iii) Existence of identity:

For $(x_1, x_2) \in \mathbf{C}$, there exists $(0, 0) \in \mathbf{C}$ such that

$$\begin{aligned}(x_1, x_2) + (0, 0) &= (x_1 + 0, x_2 + 0) \\ &= (x_1, x_2)\end{aligned}$$

Also
$$\begin{aligned}(0, 0) + (x_1, x_2) &= (0 + x_1, 0 + x_2) \\ &= (x_1, x_2)\end{aligned}$$

$$\therefore (x_1, x_2) + (0, 0) = (x_1, x_2) = (0, 0) + (x_1, x_2)$$

(iv) Existence of inverse:

For any $(x_1, x_2) \in \mathbf{C}$, there exists $(-x_1, -x_2) \in \mathbf{C}$ such that

$$(x_1, x_2) + (-x_1, -x_2) = (0, 0) = (-x_1, -x_2) + (x_1, x_2)$$

Thus $\mathbf{C}$ is an abelian group with respect to vector addition.

II. Properties of scalar multiplication in $\mathbf{C}$.

$$\begin{aligned}(i) \quad a[(x_1, x_2) + (y_1, y_2)] &= a(x_1 + y_1, x_2 + y_2) \\ &= (a(x_1 + y_1), a(x_2 + y_2)) \\ &= (ax_1 + ay_1, ax_2 + ay_2) \\ &= (ax_1, ax_2) + (ay_1, ay_2) \\ &= a(x_1, x_2) + a(y_1, y_2).\end{aligned}$$

Thus $a[(x_1, x_2) + (y_1, y_2)] = a(x_1, x_2) + a(y_1, y_2)$, for all $(x_1, x_2), (y_1, y_2) \in \mathbf{C}$, $a \in \mathbf{R}$

$$\begin{aligned}(ii) \quad (a + b)(x_1, x_2) &= ((a + b)x_1, (a + b)x_2) \\ &= (ax_1 + bx_1, ax_2 + bx_2) \\ &= (ax_1, ax_2) + (bx_1, bx_2) \\ &= a(x_1, x_2) + b(x_1, x_2)\end{aligned}$$

Thus $(a + b)(x_1, x_2) = a(x_1, x_2) + b(x_1, x_2)$, for all $(x_1, x_2) \in \mathbf{C}$ and $a, b \in \mathbf{R}$

$$\begin{aligned}(iii) \quad a(b(x_1, x_2)) &= a(bx_1, bx_2) \\ &= (abx_1, abx_2) \\ &= (ab)(x_1, x_2)\end{aligned}$$

Thus
$$a(b(x_1, x_2)) = (ab)(x_1, x_2)$$

$$(iv) \quad 1 \cdot (x_1, x_2) = (x_1, x_2) \text{ for all } (x_1, x_2) \in C$$

Thus the set C satisfies all the properties of vector space and hence C is a vector space over $\mathbf{R}$.

Example 3.

Show that the set of all matrices of the type $m \times n$ where m and n are fixed positive integers is a vector space over $\mathbf{R}$ with respect to matrix addition and multiplication of matrix by a scalar (i.e., a real number).

Solution. Let M denotes the set of all matrices of type $m \times n$.

Now, **matrix addition** is an internal binary composition in the set M .

We know that sum of two matrices of the same type $m \times n$ is always defined and is a matrix of type $m \times n$ and hence belongs to the same set M .

Also, **multiplication of a matrix** $A = [a_{ij}]_{m \times n}$ by a scalar $\alpha (\alpha \in \mathbf{R})$ is a matrix

$$\alpha A = [\alpha a_{ij}]_{m \times n}$$

and hence is an external binary composition in the set M .

I. $(M, +)$ is an abelian group.

(i) **Associativity:**

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$ and $C = [c_{ij}]_{m \times n}$ be three matrices belonging to set M .

$$\text{Now, } (A + B) + C = ([a_{ij}] + [b_{ij}]) + [c_{ij}]$$

$$= [a_{ij} + b_{ij}] + [c_{ij}]$$

$$= [(a_{ij} + b_{ij}) + (c_{ij})]$$

$$= [a_{ij} + (b_{ij} + c_{ij})]$$

[$\because$ Addition is associative in $\mathbf{R}$]

$$= [a_{ij}] + [b_{ij} + c_{ij}]$$

$$= [a_{ij}] + ([b_{ij}] + [c_{ij}])$$

$$= A + (B + C)$$

(ii) **Commutativity:**

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ be two matrices belonging to set M .

$\therefore$

$$A + B = [a_{ij}] + [b_{ij}]$$

$$= [a_{ij} + b_{ij}]$$

$$= [b_{ij} + a_{ij}]$$

[$\because$ Addition is commutative in $\mathbf{R}$]

$$= [b_{ij}]_{m \times n} + [a_{ij}]_{m \times n}$$

$$= B + A$$

(iii) Existence of Identity :

Let $A = [a_{ij}]_{m \times n} \in M$ be arbitrary.

We know that a zero matrix (null matrix) of type $m \times n$ also belongs to the set M and is denoted by O .

$$\begin{aligned}\text{Now,} \quad A + O &= [a_{ij}] + [O] = [a_{ij} + O] \\ &= [a_{ij}] = A\end{aligned}$$

$$\text{Similarly,} \quad O + A = A.$$

(iv) Existence of Inverse :

If $A = [a_{ij}]_{m \times n}$ belongs to the set M , then

$-A = [-a_{ij}]_{m \times n}$ also belongs to the set M

$$\begin{aligned}\text{Now,} \quad A + (-A) &= [a_{ij}] + [-a_{ij}] \\ &= [a_{ij} - a_{ij}] \\ &= [0] = O\end{aligned}$$

$$\text{Similarly,} \quad (-A) + A = O$$

Thus the set M is an abelian group under addition.

II. Properties of Scalar Multiplication in M .

(i) Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n} \in M$ and $a \in \mathbf{R}$, then

$$\begin{aligned}a(A + B) &= a([a_{ij}] + [b_{ij}]) \\ &= a[a_{ij} + b_{ij}] \\ &= [a(a_{ij} + b_{ij})] \\ &= [a \cdot a_{ij} + a \cdot b_{ij}] \quad [\because \text{Multiplication is distributive in } \mathbf{R}] \\ &= [a \cdot a_{ij}] + [a \cdot b_{ij}] \\ &= a[a_{ij}] + a[b_{ij}] \\ &= aA + aB.\end{aligned}$$

(ii) Let $A = [a_{ij}]_{m \times n} \in M$ and $a, b \in \mathbf{R}$, then

$$\begin{aligned}(a + b)A &= (a + b)[a_{ij}] \\ &= [(a + b)a_{ij}] \\ &= [a \cdot a_{ij} + b \cdot a_{ij}] \\ &= [a \cdot a_{ij}] + [b \cdot a_{ij}] \\ &= a[a_{ij}] + b[a_{ij}] \\ &= aA + bA\end{aligned}$$

(iii) Let $A = [a_{ij}]_{m \times n}$ belong to the set M and $a, b, \in \mathbf{R}$ then

$$\begin{aligned} a(bA) &= a(b[a_{ij}]) \\ &= a([ba_{ij}]) \\ &= [(ab)a_{ij}] \\ &= (ab)[a_{ij}] \\ &= (ab)A \end{aligned}$$

i.e.,

$$a(bA) = (ab)A$$

(iv) Let $A = [a_{ij}]_{m \times n}$ belong to the set M , then

$$1 \cdot A = 1 \cdot [a_{ij}] = [1 \cdot a_{ij}] = [a_{ij}] = A.$$

Thus M satisfies all the properties of vector space and hence M is a vector space over $\mathbf{R}$.

Example 4.

Is the set $M_{2 \times 2}$ of all 2×2 real matrices, a vector space under vector addition and scalar multiplication defined as follows :

$$A + B = A \cdot B, \text{ for all } A, B \in M_{2 \times 2}$$

and

$$a \cdot A = aA, \text{ for all } A \in \mathbf{R} \text{ and } A \in M_{2 \times 2} ?$$

Solution. This is not a vector space because it does not satisfy the property of commutative law for addition as shown by following example :

Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$ be two matrices belonging to set $M_{2 \times 2}$. Then

$$\begin{aligned} A + B = A \cdot B &\Rightarrow \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 23 & 31 \\ 34 & 46 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{Also, } B + A = B \cdot A &\Rightarrow \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 19 & 43 \\ 22 & 50 \end{bmatrix} \end{aligned}$$

Hence $A + B \neq B + A$.

Note.

It should be noted that in the vector addition defined as $A + B = A \cdot B$, the '+' symbol on L.H.S. is only a well defined operation for combining two vectors and the '.' symbol on R.H.S. is for standard matrix multiplication. So it should always be remembered that "a vector space does not require a standard vector addition but a well defined operation for combining two vectors that satisfies the properties listed in definition 1".

Example 5.

Is the set M of 2×1 column matrices, a vector space over R with respect to vector addition and scalar multiplication defined as follows :

$$[a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}],$$

$$\alpha \cdot [a_{ij}] = [-\alpha a_{ij}],$$

where $[a_{ij}]$, $[b_{ij}]$ belong to M and $\alpha \in R$?

Solution. It is clear that all the properties of vector addition hold. Also the set M is closed under the given scalar multiplication. But we notice that

$$(\alpha\beta) \cdot [a_{ij}] = (\alpha\beta) \cdot \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} -(\alpha\beta) a_{11} \\ -(\alpha\beta) a_{21} \end{bmatrix} = \begin{bmatrix} -\alpha\beta a_{11} \\ -\alpha\beta a_{21} \end{bmatrix},$$

while

$$\alpha \cdot (\beta \cdot [a_{ij}]) = \alpha \cdot \begin{bmatrix} -\beta a_{11} \\ -\beta a_{21} \end{bmatrix} = \begin{bmatrix} (-\alpha)(-\beta a_{11}) \\ (-\alpha)(-\beta a_{21}) \end{bmatrix} = \begin{bmatrix} \alpha\beta a_{11} \\ \alpha\beta a_{21} \end{bmatrix}$$

These two expressions are not equal for all non-zero α and β belonging to R .

Further,

$$1 \cdot [a_{ij}] = 1 \cdot \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} -a_{11} \\ -a_{21} \end{bmatrix} \neq [a_{ij}]$$

Thus, two of the properties for scalar multiplication do not hold, and hence the given set is not a vector space.

Note.

It should be noted that in the scalar multiplication defined as $\alpha \cdot [a_{ij}] = [-\alpha a_{ij}]$, the '.' symbol on L.H.S. is only a well defined operation for combining a scalar with a vector, regardless of how unconventional the operation may be. So it should always be remembered that "a vector space does not require a standard scalar multiplication, but a well defined process for combining scalars and vectors that satisfies the properties listed in definition 1".

Example 6.

Show that the set V of all real valued continuous functions of x defined on interval $[0, 1]$ is a vector space over the field $\mathbb{R}$ of real numbers with respect to vector addition and scalar multiplication defined by

$$(f_1 + f_2)x = f_1(x) + f_2(x), \text{ for all } f_1, f_2 \in V$$

$$(af_1)x = af_1(x), \text{ for all } a \in \mathbb{R}, f_1 \in V$$

Solution. Since sum of two continuous functions is continuous

$$\therefore f_1 + f_2 \in V \text{ for all } f_1, f_2 \in V$$

Thus the addition composition defined in V is an **internal binary composition** in V .

Again af_1 is continuous function if f_1 is continuous, and so

$$af_1 \in V \text{ for all } a \in \mathbb{R} \text{ and } f_1 \in V$$

Thus the scalar multiplication defined in V is an **external binary composition** in V over $\mathbb{R}$.

I. $(V, +)$ is an abelian group.

(i) **Associativity:**

Let $f_1, f_2, f_3 \in V$ be arbitrary

$$\begin{aligned} [(f_1 + f_2) + f_3](x) &= (f_1 + f_2)(x) + f_3(x) \\ &= [f_1(x) + f_2(x)] + f_3(x) \\ &= f_1(x) + [f_2(x) + f_3(x)] \\ &= f_1(x) + (f_2 + f_3)(x) \\ &= [f_1 + (f_2 + f_3)](x) \end{aligned}$$

$$\therefore (f_1 + f_2) + f_3 = f_1 + (f_2 + f_3)$$

(ii) **Commutativity:**

Let $f_1, f_2 \in V$ be arbitrary.

$$\begin{aligned} (f_1 + f_2)(x) &= f_1(x) + f_2(x) \\ &= f_2(x) + f_1(x) \\ &= (f_2 + f_1)(x), \text{ for all } x \end{aligned}$$

$$\therefore f_1 + f_2 = f_2 + f_1$$

(iii) **Existence of Identity:**

Define a function $\mathbf{0}$ such that $\mathbf{0}(x) = 0$ (real number), for all $x \in [0, 1]$

Also, $\mathbf{0}$ is a continuous function and belongs to V

Now,

$$\begin{aligned}(\mathbf{0} + f_1)(x) &= \mathbf{0}(x) + f_1(x) \\ &= 0 + f_1(x) = f_1(x)\end{aligned}$$

and

$$(f_1 + \mathbf{0})(x) = f_1(x) + \mathbf{0}(x) = f_1(x)$$

$$\therefore \mathbf{0} + f_1 = f_1 + \mathbf{0} = f_1$$

Thus, the function $\mathbf{0}$ defined above is an additive identity element of V .

(iv) Existence of Inverse:

For a function f_1 , the function $-f_1$ defined by

$$(-f_1)(x) = -f_1(x), \text{ is called additive inverse, as}$$

$$\begin{aligned}[f_1 + (-f_1)](x) &= f_1(x) + (-f_1)(x) \\ &= f_1(x) - f_1(x) = 0 = \mathbf{0}(x)\end{aligned}$$

$$\text{Similarly, } [-f_1 + f_1](x) = \mathbf{0}(x).$$

Thus the set V is an abelian group under addition.

II. Properties of Scalar Multiplication in V .

(i) For $f_1, f_2 \in V$ and $a \in \mathbf{R}$, we have

$$\begin{aligned}[a(f_1 + f_2)]x &= a[(f_1 + f_2)(x)] \\ &= a[f_1(x) + f_2(x)] \\ &= af_1(x) + af_2(x) \\ &= (af_1)x + (af_2)x \\ &= (af_1 + af_2)x\end{aligned}$$

$$\therefore a(f_1 + f_2) = af_1 + af_2$$

(ii) For $f_1 \in V$ and $a, b \in \mathbf{R}$

$$\begin{aligned}[(a+b)f_1](x) &= (a+b)f_1(x) \\ &= af_1(x) + bf_1(x) \\ &= (af_1)x + (bf_1)x \\ &= (af_1 + bf_1)(x)\end{aligned}$$

$$\therefore (a+b)f_1 = af_1 + bf_1$$

(iii) For $f_1 \in V$ and $a, b \in \mathbf{R}$, we have

$$\begin{aligned}[a(bf_1)]x &= a[(bf_1)x] \\ &= a[bf_1(x)] \\ &= (ab)f_1(x)\end{aligned}$$

$$= [(ab) f_1] x$$

$$\therefore a(b f_1) = (ab) f_1$$

(iv) For $f_1 \in V$ and $1 \in \mathbf{R}$, we have

$$(1 \cdot f_1)x = 1 \cdot f_1(x) = f_1(x)$$

$$\therefore 1 f_1 = f_1.$$

Thus V satisfies all the properties of a vector space and hence V is a vector space.

Example 7.

Show that the set $\mathbf{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbf{Q}\}$ is a vector space over $\mathbf{Q}$ with respect to the compositions:

$$(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c + (b + d)\sqrt{2})$$

and

$$\alpha(a + b\sqrt{2}) = \alpha a + b\alpha\sqrt{2}$$

where a, b, c, d and α are all rational numbers.

Solution. The given set is

$$\mathbf{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbf{Q}\}$$

I. Properties of Addition :

Let $(a + b\sqrt{2}), (c + d\sqrt{2}), (e + f\sqrt{2}) \in \mathbf{Q}(\sqrt{2})$ be arbitrary

(i) Associativity :

$$\begin{aligned} [(a + b\sqrt{2}) + (c + d\sqrt{2})] + (e + f\sqrt{2}) &= (a + c + (b + d)\sqrt{2}) + (e + f\sqrt{2}) \\ &= a + c + e + (b + d + f)\sqrt{2} \end{aligned}$$

$$\begin{aligned} \text{Also, } (a + b\sqrt{2}) + [(c + d\sqrt{2}) + (e + f\sqrt{2})] &= (a + b\sqrt{2}) + (c + e + (d + f)\sqrt{2}) \\ &= a + c + e + (b + d + f)\sqrt{2} \end{aligned}$$

$$\therefore [(a + b\sqrt{2}) + (c + d\sqrt{2})] + (e + f\sqrt{2}) = (a + b\sqrt{2}) + [(c + d\sqrt{2}) + (e + f\sqrt{2})]$$

(ii) Commutativity :

$$\begin{aligned} (a + b\sqrt{2}) + (c + d\sqrt{2}) &= (a + c + (b + d)\sqrt{2}) = (c + a + (d + b)\sqrt{2}) \\ &= (c + d\sqrt{2}) + (a + b\sqrt{2}) \end{aligned}$$

(iii) Existence of Identity :

There exists $\mathbf{0} = 0 + 0\sqrt{2} \in V$ such that $(a + b\sqrt{2}) + (0 + 0\sqrt{2}) = a + b\sqrt{2}$

and $(0 + 0\sqrt{2}) + (a + b\sqrt{2}) = a + b\sqrt{2}$

(iv) Existence of Inverse :

For each $a + b\sqrt{2} \in V$, there exists $(-a - b\sqrt{2}) \in V$ such that

$$(a + b\sqrt{2}) + (-a - b\sqrt{2}) = 0 + 0\sqrt{2} = 0$$

and

$$(-a - b\sqrt{2}) + (a + b\sqrt{2}) = 0 + 0\sqrt{2} = 0$$

II. Properties of Scalar Multiplication

If $a + b\sqrt{2} \in V$ and $\alpha \in \mathbb{Q}$, then

$\alpha(a + b\sqrt{2}) = \alpha a + \alpha b\sqrt{2} \in V$ defines the scalar multiplication.

Moreover, if $u = a + b\sqrt{2}, v = c + d\sqrt{2}$ where $a, b, c, d, \alpha, \beta \in \mathbb{Q}$ be arbitrary, then

$$(i) \quad \alpha(u + v) = \alpha(a + b\sqrt{2} + c + d\sqrt{2})$$

$$= \alpha(a + c + (b + d)\sqrt{2}) = \alpha(a + c) + \alpha(b + d)\sqrt{2}$$

$$= (\alpha a + \alpha b\sqrt{2}) + (\alpha c + \alpha d\sqrt{2}) = \alpha u + \alpha v$$

$$(ii) \quad (\alpha + \beta)u = (\alpha + \beta)(a + b\sqrt{2}) = (\alpha + \beta)a + (\alpha + \beta)b\sqrt{2}$$

$$= (\alpha a + \alpha b\sqrt{2}) + (\beta a + \beta b\sqrt{2}) = \alpha u + \beta u$$

$$(iii) \quad (\alpha\beta)u = (\alpha\beta)(a + b\sqrt{2}) = (\alpha\beta a + \alpha\beta b\sqrt{2})$$

$$= \alpha(\beta a + \beta b\sqrt{2}) = \alpha(\beta u).$$

$$(iv) \quad 1 \cdot u = 1 \cdot (a + b\sqrt{2}) = a + b\sqrt{2} = u$$

Hence $\mathbb{Q}(\sqrt{2})$ is a vector space over $\mathbb{Q}$.

Example 8.

Show that any field forms a vector space over itself.

Solution. Let F be any field. Since F is a field, it has two binary operations defined in it, say addition and multiplication such that

I. F forms an abelian group with respect to addition.

[$\because F$ is a field]

II. **Properties of Scalar Multiplication :** By the properties of F being a field, we have

$$(i) \quad a(u + v) = au + av, \text{ for all } a, u, v \in F$$

$$(ii) \quad (a + b)u = au + bu, \text{ for all } a, b, u \in F$$

$$(iii) \quad a(bu) = (ab)u, \text{ for all } a, b, u \in F$$

$$(iv) \quad 1u = u, \text{ for all } u \in F$$

Hence F is a vector space over itself.

Example 9.

If H is a sub field of a field F , then show that F can be considered as a vector space over H .

Solution. I. Let us take the addition in F as the internal binary operation. Thus F being a field, forms an abelian group with respect to addition

II. Also we consider the multiplication of the elements of the field F as an external binary operation over H , i.e., $a \in H, u \in F \Rightarrow au \in F$.

Moreover, F satisfies the following properties of scalar multiplication.

- (i) $a(u+v) = au + av$, for all $a \in H, u, v \in F$
- (ii) $(a+b)u = au + bu$, for all $a, b \in H, u \in F$
- (iii) $a(bu) = (ab)u$, for all $a, b \in H, u \in F$
- (iv) $1u = u$, for all $1 \in H, u \in F$

Thus F is a vector space over the subfield H .

Note.

The field $\mathbf{R}$ of all real numbers being a subfield of the field $\mathbf{C}$ (of all complex numbers) suggests that $\mathbf{C}$ is a vector space over $\mathbf{R}$. But note that $\mathbf{R}$ is not a vector space over $\mathbf{C}$, as $\mathbf{R}$ is not closed with respect to scalar multiplication. e.g., $3 \in \mathbf{R}, 4+5i \in \mathbf{C}$, but $3(4+5i) \notin \mathbf{R}$.

EXERCISE 1.1

1. Prove that

- (i) $\mathbf{C}$ is a vector space over $\mathbf{C}$
 - (ii) $\mathbf{C}$ is a vector space over $\mathbf{R}$
 - (iii) $\mathbf{Q}$ is not a vector space over $\mathbf{R}$
 - (iv) $\mathbf{R}$ is not a vector space over $\mathbf{C}$
- under usual operations of addition and scalar multiplication.

2. Which of the following sets form vector spaces over reals ?

- (i) All polynomials over $\mathbf{R}$ with constant term zero.
- (ii) All polynomials over $\mathbf{R}$ with constant term 1.
- (iii) Set of all ordered pairs (a, b) of integers.
- (iv) All polynomials with positive coefficients

3. Let $\mathbf{R}^+$ be the set of all positive real numbers. Define the operations of addition and scalar multiplication as follows :

$$u+v = uv, \text{ for all } u, v \in \mathbf{R}^+$$

$$au = u^a, \text{ for all } u \in \mathbf{R}^+ \text{ and } a \in \mathbf{R}.$$

Show that $\mathbf{R}^+$ is a vector space over $\mathbf{R}$.

4. Show that $\mathbf{R}$ is a vector space over the field $\mathbf{Q}$ of rationals where vector addition is defined by

$$u + v = u + v, \text{ for all } u, v \in \mathbf{R}$$

and scalar multiplication is defined by :

$$a \cdot u = au, \text{ where } a \in \mathbf{Q}, u \in \mathbf{R}$$

5. Which of the following subsets of $\mathbf{R}^4$ are vector spaces for co-ordinatewise addition and scalar multiplication ?

The set of all vectors $(x_1, x_2, x_3, x_4) \in \mathbf{R}^4$ such that

- (a) $x_1 = 0$ (b) $x_1 < 0$ (c) $x_2 > 0$
(d) $x_4 = 0$ (e) $x_3^2 \geq 0$ (f) $2x_1 + 3x_2 = 0$.

6. Consider the set $\mathbf{R}^n$ of all ordered n -tuples of real numbers defined by :

$$\mathbf{R}^n = \{u = (x_1, x_2, \dots, x_n) : x_i \text{ is real, } i = 1, \dots, n\}$$

Show that $\mathbf{R}^n$ is a vector space over $\mathbf{R}$.

7. Prove that the set of all matrices of the form $\begin{bmatrix} x & y \\ -y & x \end{bmatrix}$, where $x, y \in \mathbf{C}$, is a vector space over $\mathbf{C}$ with respect to matrix addition and scalar multiplication.
8. Prove that the set of all diagonal matrices over $\mathbf{R}$, is a vector space with respect to matrix addition and scalar multiplication.
9. Let $V = \{(a, b) : a, b \in \mathbf{R}\}$ and $F = \mathbf{R}$. Show that V is not a vector space under addition and scalar multiplication defined as in each one of the following cases :
- (i) $(a, b) + (c, d) = (a + c, b + d)$; $\alpha(a, b) = (0, \alpha b)$
(ii) $(a, b) + (c, d) = (0, b + d)$; $\alpha(a, b) = (\alpha a, \alpha b)$
(iii) $(a, b) + (c, d) = (a + c, b + d)$; $\alpha(a, b) = (\alpha a, b)$

ANSWERS

2. (i) Yes (ii) No, not closed under addition
(iii) No, not closed under scalar multiplication
(iv) No, not closed under scalar multiplication

5. (a), (d), (e), (f)

1.4. SOME GENERAL PROPERTIES OF VECTOR SPACES

In any vector space $V(F)$, the following results hold :

- (i) $a\mathbf{0} = \mathbf{0}$, for all $a \in F$
- (ii) $0u = \mathbf{0}$, for all $u \in V$
- (iii) $a(-u) = -(au) = (-a)u$, for all $a \in F$ and $u \in V$
- (iv) $a(u - v) = au - av$, for all $a \in F$ and $u, v \in V$
- (v) $au = \mathbf{0} \Rightarrow a = 0$ or $u = \mathbf{0}$.
- (vi) $au = bu$ and $u \neq \mathbf{0} \Rightarrow a = b$
- (vii) $au = av$ and $a \neq 0 \Rightarrow u = v$

Proof. (i) Since $a\mathbf{0} = a(\mathbf{0} + \mathbf{0}) = a\mathbf{0} + a\mathbf{0}$

$$\Rightarrow \mathbf{0} + a\mathbf{0} = a\mathbf{0} + a\mathbf{0} \quad [\because \mathbf{0} + a\mathbf{0} = a\mathbf{0} \in V]$$

$$\Rightarrow \mathbf{0} = a\mathbf{0} \quad [\text{Cancellation in } V]$$

or

$$a\mathbf{0} = \mathbf{0}$$

(ii) Since $\mathbf{0} + \mathbf{0} = \mathbf{0}$

$$\Rightarrow (\mathbf{0} + \mathbf{0})u = \mathbf{0}u$$

$$\Rightarrow 0u + 0u = 0u$$

$$\Rightarrow 0u + 0u = 0u + \mathbf{0}$$

$$\Rightarrow 0u = \mathbf{0} \quad [\text{Cancellation in } V]$$

(iii) Since $a[u + (-u)] = au + a(-u)$

$$\Rightarrow a\mathbf{0} = au + a(-u)$$

$$\Rightarrow \mathbf{0} = au + a(-u)$$

$$\Rightarrow a(-u) \text{ is additive inverse of } au$$

$$\Rightarrow a(-u) = -(au)$$

Also $a(bu) = ab(u)$ for all $a, b \in F$ and $u \in V$

$$\therefore -(au) = -a(u)$$

$$\text{Thus } a(-u) = -(au) = -a(u)$$

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$$\begin{aligned}
 (iv) \quad a(u-v) &= a[u + (-v)] \\
 &= au + a(-v) \\
 &= au + [(-av)] \\
 &= au - av
 \end{aligned}$$

(v) Let $au = 0$ and $a \neq 0$

Now $a \neq 0 \Rightarrow a^{-1}$ exists

$$\begin{aligned}
 \text{and} \quad a^{-1}(au) &= a^{-1}0 \\
 \Rightarrow (a^{-1}a)u &= 0 \\
 \Rightarrow 1u &= 0 \\
 \Rightarrow u &= 0
 \end{aligned}$$

Also if $au = 0$ and $u \neq 0$; we have to show that $a = 0$

Let $a \neq 0 \Rightarrow a^{-1}$ exists

$$\begin{aligned}
 \therefore a^{-1}(au) &= (a^{-1}a)u = 0 \\
 \Rightarrow 1u &= 0 \\
 \Rightarrow u &= 0
 \end{aligned}$$

which is a contradiction. Therefore a must be equal to 0.

Thus $au = 0$ and $u \neq 0 \Rightarrow a = 0$.

$$\begin{aligned}
 (vi) \quad au &= bu \\
 \Rightarrow au - bu &= 0 \\
 \Rightarrow (a-b)u &= 0
 \end{aligned}$$

Since $u \neq 0$

$$\begin{aligned}
 \Rightarrow a - b &= 0 \\
 \Rightarrow a &= b
 \end{aligned}$$

$$(vii) \quad au = av$$

$$\begin{aligned}
 \Rightarrow au - av &= 0 \\
 \Rightarrow a(u-v) &= 0 \\
 \Rightarrow u - v &= 0 \\
 \Rightarrow u &= v,
 \end{aligned}$$

$[\because a \neq 0]$

1.5. SUBSPACES

As usual, in the study of any algebraic structure, it is of interest to study subsets which have the same structure as the set under consideration. In this section we introduce the appropriate notion of substructure for a vector space.

Definition : A non-empty subset W of a vector space $V(F)$ is said to form a **subspace** of V if W is also a vector space over F with the same addition and scalar multiplication as for V .

Examples :

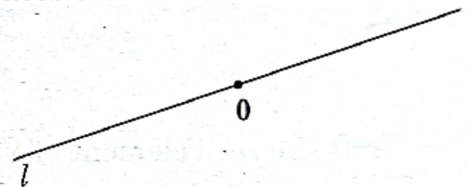
(1) Let $W_1 = \{(a, 0, 0) : a \in \mathbf{R}\}$

and $W_2 = \{(a, b, 0) : a, b \in \mathbf{R}\}.$

Here W_1 is a subspace of W_2 . Also W_1 and W_2 are subspaces of $\mathbf{R}^3$.

(2) The entire space V and the zero space $\{0\}$ are subspaces of V and are called **trivial subspaces** of V .

(3) Each line l in $\mathbf{R}^n$ that passes through 0 is a subspace of $\mathbf{R}^n$.



Remark.

1. The zero space (null space) $\{0\}$ consisting of the single element zero vector and the vector space V itself are subspaces of every vector space $V(F)$ and are called **improper subspaces**. All other subspaces, if any, are called **proper subspaces**.
2. To show that a subset W of a vector space V is a subspace of V , we must verify that definition holds for W . But the eight properties required for a vector space are sure to be true for the subset W , because they hold in all of V .

Thus, instead of verifying eight properties, we need to verify only the two closure properties, i.e. closure under addition and closure under scalar multiplication.

1.6. SOME THEOREMS ON VECTOR SUB-SPACES

1.6.1. Theorem. A necessary and sufficient condition for a non-empty subset W of a vector space $V(F)$ to be a subspace of V is that W is closed under addition and scalar multiplication. [K.U. 2008]

Proof. The condition is necessary :

If W is a subspace of V , then by definition, W is a vector space over F and thus the conditions

$$u, v \in W \Rightarrow u + v \in W$$

and

$$u \in W, a \in F \Rightarrow au \in W \quad \text{hold good.}$$

Therefore W is closed under addition and scalar multiplication.

The condition is sufficient :

Let W be closed under addition and scalar multiplication, i.e.,

$$(i) \quad u, v \in W \Rightarrow u + v \in W \quad \text{and}$$

$$(ii) \quad u \in W, a \in F \Rightarrow au \in W.$$

Since 1 is the multiplicative identity of F , so $-1 \in F$

$$\therefore (-1)u \in W$$

[By (ii)]

$$\Rightarrow -u \in W$$

$\Rightarrow$ The additive inverse of each $u \in W$ is also in W .

$$\text{Also} \quad u \in W, -u \in W \Rightarrow u + (-u) \in W$$

[By (i)]

$$\Rightarrow 0 \in W$$

Thus the zero element of V is also the zero element of W .

$$\text{Also} \quad u, v \in W \Rightarrow u, v \in V$$

and so

$$u + v = v + u \quad [\because \langle V, + \rangle \text{ is an abelian group}]$$

Similarly the condition of associativity and other conditions of a vector space also hold in W since they hold in V .

$\therefore (W, +)$ is an abelian group. Also, W is a subset of V .

Hence, W is a subspace of V .

1.6.2. Theorem. A non-empty subset W of a vector space $V(F)$ is a subspace of V if and only if $au + v \in W$ for each $a \in F$ and $u, v \in W$.

Proof. The condition is necessary :

Let W be a subspace of V . Then W is closed under addition and scalar multiplication.

Let $u, v \in W$.

$$\text{Therefore} \quad a \in F \Rightarrow au \in W$$

$$\text{Again,} \quad au \in W, v \in W \Rightarrow au + v \in W.$$

The condition is sufficient :

Let W be a non-empty subset of V such that $au + v \in W$ for each $a \in F$ and $u, v \in W$. Then :

$$(i) \quad \text{Since} \quad 1 \in F \Rightarrow -1 \in F$$

Thus taking $v = u = w$, where $w \in W$ and $a = -1$ in the given condition, we get

$$(-1)w + w = -(1w) + w = -w + w = \mathbf{0}, \text{ is the zero element of } W.$$

(ii) Again $a \in F$, $u \in W$. Then taking $v = \mathbf{0}$ in the given condition, we get $au + \mathbf{0} = au \in W$. Thus W is closed under scalar multiplication.

(iii) Let $u \in W$. Also $-1 \in F$ and W is closed under scalar multiplication.

Thus
$$(-1)u = -(1u) = -u \in W$$

(iv) Since $1 \in F$, therefore taking $a = 1$ in the given condition we see that

$$1u + v = u + v \in W \text{ for } u, v \in W.$$

Thus W is closed under vector addition.

Moreover rest of the conditions of a vector space will hold in W since they hold in V and also W is a subset of V .

Hence, W is a subspace of V .

1.6.3. Theorem. The necessary and sufficient conditions for a non-empty subset W of a vector space $V(F)$ to be a subspace are

(i) $u - v \in W$ for $u, v \in W$

(ii) $au \in W$ for $a \in F, u \in W$

Proof. The conditions are necessary :

Let W be a subspace of $V(F)$. We have to prove that conditions (i) and (ii) hold.

Since W is a subspace of V , therefore W is also a vector space over F and so condition (ii) holds.

Now,
$$v \in W \Rightarrow -v \in W$$

Again,
$$u, -v \in W \Rightarrow u + (-v) \in W$$

Thus
$$u, v \in W \Rightarrow u - v \in W \text{ and so the condition (i) also holds}$$

The conditions are sufficient :

Let W be a non-empty subset of V such that the conditions (i) and (ii) hold.

Let $u \in W$

Then by condition (i), $u - u \in W$

$$\Rightarrow \mathbf{0} \in W$$

Thus the identity element $\mathbf{0}$ of V also belongs to W .

Now by condition (ii), $-1 \in F$ and $u \in W$

$$\Rightarrow -1u \in W$$

$$\Rightarrow -u \in W \quad (\text{Existence of additive inverse})$$

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Again $u \in W, v \in W \Rightarrow u \in W, -v \in W$

$\Rightarrow u - (-v) \in W$

[By condition (i)]

$\Rightarrow u + v \in W$ for $u, v \in W$

Thus W is closed under vector addition

Also, $u, v \in W$

$\Rightarrow u, v \in V$

and so

$$u + v = v + u$$

[$\because \langle V, + \rangle$ is an abelian group]

Similarly conditions of a vector space also hold in W since they hold in V and also W is a subset of V .

Hence, W is a subspace of V .

1.64. Theorem. A non-empty subset W of a vector space $V(F)$ is a subspace of V if and only if $au + bv \in W$ for $a, b \in F$ and $u, v \in W$.

[M.D.U. 2013, 09, 07, 05; K.U. 2013]

Proof. The condition is necessary :

Let W be a subspace of $V(F)$. Then by definition W is also closed under vector addition and scalar multiplication.

Since for any $a \in F, u \in W \Rightarrow au \in W$

and for any $b \in F, v \in W \Rightarrow bv \in W$

Therefore for any $a, b \in F$ and $u, v \in W$

$$au + bv \in W.$$

The condition is sufficient :

Let W be a non-empty subset of $V(F)$ satisfying the condition that for $a, b \in F$ and $u, v \in W$,

$$au + bv \in W$$

Since $1 \in F$, so taking $a = 1, b = 1$, we get from given condition that

$$1u + 1v \in W$$

$\Rightarrow u + v \in W$ for $u, v \in W$

$\Rightarrow W$ is closed under vector addition.

Again $a \in F, u \in W \Rightarrow au = au + 0v$ for any $v \in W, 0 \in F$.

$\Rightarrow au = au + 0v \in W$

Thus $a \in F, u \in W \Rightarrow au \in W$

$\Rightarrow W$ is closed under scalar multiplication.

Hence, W is a subspace of $V(F)$.

1.6.5. Theorem. The intersection of any family of subspaces of a vector space $V(F)$ is a subspace of $V(F)$.

Proof. Let $W_1, W_2, \dots, W_r, \dots$, be subspaces of a vector space $V(F)$.

Let $W = (\cap W_r)$ be the intersection of given family of subspaces of $V(F)$.

Since $0 \in W_r$ for all $r \Rightarrow 0 \in (\cap W_r) = W$ and so $W \neq \emptyset$.

Since W_r is a subspace of V , therefore

$$a, b \in F \text{ and } u, v \in W_r \Rightarrow au + bv \in W_r \text{ for all } r \quad \dots(1)$$

Let $a, b \in F$ and $u, v \in W$

$$\Rightarrow a, b \in F \text{ and } u, v \in \cap W_r \text{ for all } r$$

$$\Rightarrow a, b \in F \text{ and } u, v \in W_r \text{ for all } r$$

$$\Rightarrow au + bv \in W_r \text{ for all } r \quad [Using (1)]$$

$$\Rightarrow au + bv \in \cap W_r = W$$

Thus $a, b \in F$ and $u, v \in \cap W_r \Rightarrow au + bv \in \cap W_r$

Hence $\cap W_r$ is a subspace of $V(F)$.

1.6.6. Theorem. The intersection of two subspaces W_1 and W_2 of a vector space $V(F)$ is also a subspace of $V(F)$. [M.D.U. 2006, 02]

Proof. Let W_1 and W_2 be two subspaces of a vector space $V(F)$.

Since $0 \in W_1$ and W_2 , therefore $W_1 \cap W_2 \neq \emptyset$.

Let $u, v \in W_1 \cap W_2$ and $a, b \in F$.

$$\text{Now } u \in W_1 \cap W_2 \Rightarrow u \in W_1 \text{ and } u \in W_2$$

$$\text{and } v \in W_1 \cap W_2 \Rightarrow v \in W_1 \text{ and } v \in W_2$$

Since W_1 and W_2 are subspaces of V ,

$$\therefore a, b \in F \text{ and } u, v \in W_1 \Rightarrow au + bv \in W_1$$

$$\text{Similarly } a, b \in F \text{ and } u, v \in W_2 \Rightarrow au + bv \in W_2$$

$$\therefore au + bv \in W_1 \cap W_2$$

$$\text{Thus } a, b \in F \text{ and } u, v \in W_1 \cap W_2 \Rightarrow au + bv \in W_1 \cap W_2$$

Hence, $W_1 \cap W_2$ is a subspace of $V(F)$.

Remark.

The union of two subspaces of a vector space $V_3(\mathbf{R})$ may not be a subspace of $V_3(\mathbf{R})$. [M.D.U. 2012, 02]

Proof. Let $W_1 = \{(x, 0, 0) : x \in \mathbf{R}\}$

and $W_2 = \{(0, y, 0) : y \in \mathbf{R}\}$ be the subspaces of the vector space $V_3(\mathbf{R})$.

Let $u, v \in W_1 \cup W_2$, so that

$$u \in W_1, v \in W_2$$

$$\Rightarrow u = (x, 0, 0), v = (0, y, 0), \text{ where } x, y \in \mathbf{R}.$$

$$\therefore W_1 \cup W_2 = \{w : w = (x, 0, 0) \text{ or } w = (0, y, 0) \text{ for all } x, y \in \mathbf{R}\}$$

Now $u, v \in W_1 \cup W_2$

$$\begin{aligned} \Rightarrow u + v &= (x, 0, 0) + (0, y, 0) \\ &= (x, y, 0) \end{aligned}$$

We note that $(x, y, 0) \notin W_1$ and W_2 both.

Therefore $u + v \notin W_1 \cup W_2 \Rightarrow W_1 \cup W_2$ is not closed under vector addition.

Hence, $W_1 \cup W_2$ is not a subspace of $V_3(\mathbf{R})$.

1.6.7. Theorem. The union of two subspaces is a subspace if and only if one is contained in the other. [M.D.U. 2012, 06]

Proof. Let W_1 and W_2 be two subspaces of a vector space $V(F)$.

Let $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$

Since W_2 is a subspace of $V(F) \Rightarrow W_1 \cup W_2$ is a subspace of $V(F)$.

Again if $W_2 \subseteq W_1$, then $W_1 \cup W_2 = W_1$

Since W_1 is a subspace of $V(F) \Rightarrow W_1 \cup W_2$ is a subspace of $V(F)$

Conversely, let us suppose that $W_1 \cup W_2$ is a subspace of $V(F)$.

We have to prove that $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

If possible, let us assume that W_1 is not a subset of W_2 and W_2 is not a subset of W_1 .

Now $W_1 \not\subseteq W_2 \Rightarrow$ there exist $u \in W_1$ and $u \notin W_2$... (1)

Also $W_2 \not\subseteq W_1 \Rightarrow$ there exist $v \in W_2$ and $v \notin W_1$... (2)

From (1) and (2), it follows that $u \in W_1 \cup W_2$ and $v \in W_1 \cup W_2$.

Since $W_1 \cup W_2$ is a subspace of $V(F)$

$$\Rightarrow u + v \in W_1 \cup W_2$$

$$\Rightarrow u + v \in W_1 \text{ or } u + v \in W_2$$

If $u + v \in W_1$, then since $u \in W_1$, therefore

$$(u + v) - u \in W_1 \quad [\text{Since } W_1 \text{ is a subspace}]$$

$$\Rightarrow v \in W_1$$

But from (2), we have $v \notin W_1$, which is a contradiction.

Again if $u + v \in W_2$, then since $v \in W_2$, therefore

$$(u + v) - v \in W_2$$

$$\Rightarrow u \in W_2$$

But from (1), we have $u \notin W_2$ and so there is again a contradiction.

Thus our assumption, i.e., $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$ is wrong.

Hence, it follows that either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

SOLVED EXAMPLES

Example 1.

Let R be the field of real numbers. Which of the following are sub-spaces of $V_3(R)$

(i) $\{(x, 2y, 3z) : x, y, z \in R\}$

(ii) $\{(x, x, x) : x \in R\}$

(iii) $\{(x, y, z) : x, y, z \text{ are rational numbers}\}$

[M.D.U. 2005]

Solution. (i) Let $W = \{(x, 2y, 3z) : x, y, z \in R\}$

Let $u = (x_1, 2y_1, 3z_1)$ and $v = (x_2, 2y_2, 3z_2)$ be two elements of W

If $a, b \in R$, then

$$\begin{aligned} au + bv &= a(x_1, 2y_1, 3z_1) + b(x_2, 2y_2, 3z_2) \\ &= (ax_1, 2ay_1, 3az_1) + (bx_2, 2by_2, 3bz_2) \\ &= (ax_1 + bx_2, 2ay_1 + 2by_2, 3az_1 + 3bz_2) \\ &= (x', 2y', 3z') \text{ (say)} \end{aligned}$$

where $x' = ax_1 + bx_2$, $y' = ay_1 + by_2$ and $z' = az_1 + bz_2 \in R$

$$\therefore a, b \in \mathbf{R} \text{ and } u, v \in W \Rightarrow au + bv \in W$$

Thus, W is a sub-space of $V_3(\mathbf{R})$.

(ii) Let $W = \{(x, x, x) : x \in \mathbf{R}\}$

Let $u = (x_1, x_1, x_1)$ and $v = (x_2, x_2, x_2) \in W$ be arbitrary.

If $a, b \in \mathbf{R}$, then

$$\begin{aligned} au + bv &= a(x_1, x_1, x_1) + b(x_2, x_2, x_2) \\ &= (ax_1, ax_1, ax_1) + (bx_2, bx_2, bx_2) \\ &= (ax_1 + bx_2, ax_1 + bx_2, ax_1 + bx_2) \\ &= (x', x', x'), \text{ where } x' = ax_1 + bx_2 \in \mathbf{R} \end{aligned}$$

$$\therefore a, b \in \mathbf{R} \text{ and } u, v \in W \Rightarrow au + bv \in W$$

$\therefore W$ is a sub-space of $V_3(\mathbf{R})$.

(iii) Let $W = \{(x, y, z) : x, y, z \text{ are rational numbers}\}$

Consider an element $u = (2, 3, 6)$ of W .

Also let $a = \sqrt{2} \in \mathbf{R}$

$$\begin{aligned} \text{Then, } au &= \sqrt{2}(2, 3, 6) \\ &= (2\sqrt{2}, 3\sqrt{2}, 6\sqrt{2}) \\ &\notin W \end{aligned}$$

$[\because 2\sqrt{2}, 3\sqrt{2}, 6\sqrt{2} \text{ are not rationals}]$

$\therefore W$ is not closed under scalar multiplication.

Hence, W is not a sub-space of $V_3(\mathbf{R})$.

Example 2.

Let $V = \mathbf{R}^3$ be the Euclidean 3-space. Let

$W = \{(x, y, z) : ax + by + cz = 0; x, y, z \in \mathbf{R}\}$, a, b, c being fixed real numbers.

Show that W is a subspace of V .

OR

Show that any plane passing through the origin is a subspace of $\mathbf{R}^3$

Solution. Let $u = (x_1, y_1, z_1)$, $v = (x_2, y_2, z_2)$ be any two elements of W where $x_1, x_2, y_1, y_2, z_1, z_2 \in \mathbf{R}$.

$$\text{Then } ax_1 + by_1 + cz_1 = 0 \text{ and } ax_2 + by_2 + cz_2 = 0 \quad \dots(1)$$

$$\text{For } \alpha \in \mathbf{R}, \quad \alpha u + v = \alpha(x_1, y_1, z_1) + (x_2, y_2, z_2)$$

$$\begin{aligned}
 &= (\alpha x_1, \alpha y_1, \alpha z_1) + (x_2, y_2, z_2) \\
 &= (\alpha x_1 + x_2, \alpha y_1 + y_2, \alpha z_1 + z_2) \quad \dots(2)
 \end{aligned}$$

where $\alpha x_1 + x_2, \alpha y_1 + y_2, \alpha z_1 + z_2 \in \mathbf{R}$

$$\begin{aligned}
 \text{Now, } a(\alpha x_1 + x_2) + b(\alpha y_1 + y_2) + c(\alpha z_1 + z_2) &= a(ax_1 + by_1 + cz_1) + (ax_2 + by_2 + cz_2) \\
 &= a(0) + 0 = 0 \quad \dots(3) \quad [By (1)]
 \end{aligned}$$

From (2) and (3), we have $\alpha u + v \in W$

Thus, W is a subspace of V .

Example 3.

Let V be the vector space of all square matrices over $\mathbf{R}$. Determine which of the following are sub-spaces of V .

$$\begin{aligned}
 (i) \quad W &= \left\{ \begin{bmatrix} x & y \\ z & 0 \end{bmatrix} : x, y, z \in \mathbf{R} \right\} & (ii) \quad W &= \left\{ \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} : x, y \in \mathbf{R} \right\} \\
 (iii) \quad W &= \{A : A \in V \text{ and } A \text{ is singular}\} & (iv) \quad W &= \{A : A \in V, A^2 = A\}
 \end{aligned}$$

Solution. (i) Let

$$A = \begin{bmatrix} x_1 & y_1 \\ z_1 & 0 \end{bmatrix}$$

and

$$B = \begin{bmatrix} x_2 & y_2 \\ z_2 & 0 \end{bmatrix} \text{ be any two elements of } W.$$

If $a, b \in \mathbf{R}$, then

$$\begin{aligned}
 aA + bB &= a \begin{bmatrix} x_1 & y_1 \\ z_1 & 0 \end{bmatrix} + b \begin{bmatrix} x_2 & y_2 \\ z_2 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} ax_1 & ay_1 \\ az_1 & 0 \end{bmatrix} + \begin{bmatrix} bx_2 & by_2 \\ bz_2 & 0 \end{bmatrix} = \begin{bmatrix} ax_1 + bx_2 & ay_1 + by_2 \\ az_1 + bz_2 & 0 \end{bmatrix}
 \end{aligned}$$

which is a matrix of the type $\begin{bmatrix} x & y \\ z & 0 \end{bmatrix}$ and $ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2 \in \mathbf{R}$

$$\therefore aA + bB \in W$$

Thus, W is a sub-space of V .

(ii) Let

$$A = \begin{bmatrix} x_1 & 0 \\ 0 & y_1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} x_2 & 0 \\ 0 & y_2 \end{bmatrix} \text{ be any two elements of } W.$$

If $a, b \in \mathbf{R}$, then

$$\begin{aligned} aA + bB &= a \begin{bmatrix} x_1 & 0 \\ 0 & y_1 \end{bmatrix} + b \begin{bmatrix} x_2 & 0 \\ 0 & y_2 \end{bmatrix} \\ &= \begin{bmatrix} ax_1 + bx_2 & 0 \\ 0 & ay_1 + by_2 \end{bmatrix} \end{aligned}$$

which is a matrix of the type $\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix}$ and $ax_1 + bx_2, ay_1 + by_2 \in \mathbf{R}$

$$\therefore aA + bB \in W$$

Thus W is a sub-space of V .

(iii) Here W is the set of singular matrices.

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ be two square matrices

Since $|A| = 0$ and $|B| = 0$, therefore $A, B \in W$.

If $a, b \in \mathbf{R}$ are non zero, then

$$\begin{aligned} aA + bB &= a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & b \end{bmatrix} \\ &= \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \end{aligned}$$

Also, $|aA_1 + bA_2| = \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab \neq 0$, as none of a and b is zero.

$$\therefore aA_1 + bA_2 \notin W$$

Thus, W is not a sub-space of V .

(iv) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ so that $A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = A$

$$\therefore A \in W.$$

Now,
$$A + A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

Also,
$$(A + A)^2 = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\neq A + A$$

$\therefore (A + A)$ is not a member of the set W .

$\Rightarrow W$ is not closed under addition.

Hence, W is not a sub-space of V .

Example 4.

Let V be the vector space of all real valued continuous functions over $\mathbb{R}$. Show that the set W of solutions of differential equations $5 \frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 2y = 0$ is a subspace of V .

Solution. Let $y_1, y_2 \in W$. Then y_1, y_2 are solutions of differential equations

$$5 \frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 2y = 0 \quad \dots(1)$$

i.e.,
$$5 \frac{d^2 y_1}{dx^2} - 7 \frac{dy_1}{dx} + 2y_1 = 0 \quad \dots(2)$$

and
$$5 \frac{d^2 y_2}{dx^2} - 7 \frac{dy_2}{dx} + 2y_2 = 0 \quad \dots(3)$$

Let $a, b \in \mathbb{R}$ be arbitrary.

Now,
$$5 \frac{d^2}{dx^2} (ay_1 + by_2) - 7 \frac{d}{dx} (ay_1 + by_2) + 2(ay_1 + by_2)$$

$$= 5 \frac{d^2}{dx^2} (ay_1) - 7 \frac{d}{dx} (ay_1) + 2(ay_1) + 5 \frac{d^2}{dx^2} (by_2) - 7 \frac{d}{dx} (by_2) + 2(by_2)$$

$$= a \left[5 \frac{d^2 y_1}{dx^2} - 7 \frac{dy_1}{dx} + 2y_1 \right] + b \left[5 \frac{d^2 y_2}{dx^2} - 7 \frac{dy_2}{dx} + 2y_2 \right]$$

$$= a \cdot 0 + b \cdot 0$$

[By (2) and (3)]

$$= 0$$

Thus $ay_1 + by_2$ is a solution of (1) and so $ay_1 + by_2 \in W$.

Hence, W is a subspace of V .

EXERCISE 1.2

1. Let V be a vector space of $\mathbf{R}^3$. Examine, whether the following are subspaces of V or not :

- (i) $W = \{(x, y, z) : 3x + y - z = 0, x, y, z \in \mathbf{R}\}$
- (ii) $W = \{(x, y, z) : x + y - z = 0, 2x + 3y - z = 0, x, y, z \in \mathbf{R}\}$
- (iii) $W = \{(x_1, x_2, x_3) : x_1 + x_2 \geq 0; x_1, x_2, x_3 \in \mathbf{R}\}$
- (iv) $W = \{(x_1, x_2, x_3) : x_1 x_2 = 0; x_1, x_2, x_3 \in \mathbf{R}\}$

2. Let V be a vector space over $\mathbf{R}^3$. Examine whether the following are sub-spaces of V or not:

- (i) $W = \{(x, y, z) : x, y, z \in \mathbf{R} \text{ and } x^2 + y^2 + z^2 \leq 1\}$
- (ii) $W = \{(x, y, z) : x \geq 0, x, y, z \in \mathbf{R}\}$ [K.U. 2008]

3. (a) Let $V = \mathbf{R}^3 = \{(x, y, z) : x, y, z \in \mathbf{R}\}$. Show that the following subsets of V are subspaces of V .

- (i) $W = \{(x, y, z) : x - 3y + 4z = 0, x, y, z \in \mathbf{R}\}$
- (ii) $W = \{(x, 2y, 3z) : x, y, z \in \mathbf{R}\}$ [M.D.U. 2000]
- (iii) $W = \{(x, x, x) : x \in \mathbf{R}\}$
- (iv) $W = \{(x, y, z) : x, y, z \in \mathbf{Q}\}$

(b) Let V be the vector space of all polynomials in x of degree ≤ 5 (including the zero polynomial) over the field $\mathbf{Q}$. If $f(x) \in V$, explain which of the following are subspaces of V :

- (i) $W = \{f(x) : \text{coeff. of } x^3 \text{ is zero}\}$
- (ii) $W = \{f(x) : f(x) \text{ is a polynomial of degree } \leq 3 \text{ or } f(x) \text{ is a zero polynomial}\}$
- (iii) $W = \{f(x) : f(3) = 0\}$
- (iv) $W = \{f(x) : f(1) = 0, f(5) = 0\}$.

4. If V is the vector space of all real valued functions of real variable defined on $[0, 1]$, show that the following subsets of V are sub-spaces :

- (i) all even functions
- (ii) all continuous functions

5. If V is the vector space of all n -square matrices over $\mathbf{R}$, which of the following sets of matrices are subspaces of V ?

- (i) all lower triangular matrices
- (ii) all scalar matrices
- (iii) all symmetric matrices
- (iv) all singular matrices
- (v) all diagonal matrices

- 6/ Show that $W = \left\{ \begin{bmatrix} x & 0 \\ 0 & 0 \end{bmatrix} : x \in \mathbf{R} \right\}$ is a subspace of the vector space V , of all 2×2 real matrices.
- 7/ Show that $W = \{(x, y) : ax + by = 0; x, y \in \mathbf{R}\}$, where a, b are fixed real numbers, is a subspace of $\mathbf{R}^2$.
- 8/ Let $\mathbf{R}$ be the field of real numbers. Show that $\{(a, b, c, d) : b - 2c + d = 0\}$ is a proper subspace of $\mathbf{R}^4$ over field $\mathbf{R}$.

ANSWERS

- | | |
|--|------------------------|
| 1. (i) subspace | (ii) subspace |
| (iii) not a subspace | (iv) not a subspace |
| 2. (i) not a subspace | (ii) not a subspace |
| 3. (b) (i) subspace | (ii) subspace |
| (iii) subspace | (iv) subspace |
| 5. (i), (ii), (iii), (v) are subspaces | (iv) is not a subspace |

1.7. LINEAR SUM OF SUBSPACES

Definition : If W_1 and W_2 are two subspaces of a vector space $V(F)$, then we define the linear sum of W_1 and W_2 as

$$W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1 \text{ and } w_2 \in W_2\} \quad [M.D.U. 2003]$$

Remark,

Let $w_1 \in W_1$

Since $0 \in W_2$, therefore $w_1 = w_1 + 0 \in W_1 + W_2$

Thus $W_1 \subseteq W_1 + W_2$

Similarly $W_2 \subseteq W_1 + W_2$

Hence, $W_1 \cup W_2 \subseteq W_1 + W_2$.

1.7.1. Theorem. The linear sum of two subspaces W_1 and W_2 is also a subspace of the vector space $V(F)$.

Proof. Let W_1 and W_2 be two subspaces of a vector space $V(F)$. Let u, v be any two elements of $W_1 + W_2$. Then by definition

$$u = w_1 + w_2 \text{ for some } w_1 \in W_1, w_2 \in W_2$$

and

$$v = w_1' + w_2', \text{ where } w_1' \in W_1, w_2' \in W_2$$

$$\begin{aligned} \text{Now, } a, b \in F \Rightarrow au + bv &= a(w_1 + w_2) + b(w_1' + w_2') \\ &= (aw_1 + bw_1') + (aw_2 + bw_2') \end{aligned}$$

Since W_1 and W_2 are subspaces of V ,

$$\therefore aw_1 + bw_1' \in W_1 \text{ and } aw_2 + bw_2' \in W_2$$

Thus

$$au + bv = (aw_1 + bw_1') + (aw_2 + bw_2') \in W_1 + W_2$$

i.e.,

$$au + bv \in W_1 + W_2$$

Hence, $W_1 + W_2$ is a subspace of $V(F)$.

1.7.2. Theorem. The linear sum of two subspaces W_1 and W_2 of a vector space $V(F)$ is a subspace generated by the union of W_1 and W_2 , i.e., $W_1 + W_2 = \langle W_1 \cup W_2 \rangle$

[M.D.U. 2002]

Proof. Obviously, $W_1 \subseteq W_1 + W_2$ and $W_2 \subseteq W_1 + W_2$

$\Rightarrow$

$$W_1 \cup W_2 \subseteq W_1 + W_2$$

Since $\langle W_1 \cup W_2 \rangle$ is the smallest subspace of V containing $W_1 \cup W_2$ and $W_1 + W_2$ is a subspace containing $W_1 \cup W_2$, therefore

$$\langle W_1 \cup W_2 \rangle \subseteq W_1 + W_2 \quad \dots(1)$$

Let $u = w_1 + w_2$ be any element of $W_1 + W_2$, where $w_1 \in W_1$ and $w_2 \in W_2$

$\therefore$

$$w_1, w_2 \in W_1 \cup W_2$$

Now,

$$w_1 + w_2 = 1w_1 + 1w_2$$

$\Rightarrow$

$u = w_1 + w_2$ is a linear combination of the elements $w_1, w_2 \in W_1 \cup W_2$

$\Rightarrow$

$$u \in \langle W_1 \cup W_2 \rangle$$

$\therefore$

$$W_1 + W_2 \subseteq \langle W_1 \cup W_2 \rangle \quad \dots(2)$$

Thus from (1) and (2), we get the desired result,

i.e.,

$$W_1 + W_2 = \langle W_1 \cup W_2 \rangle$$

1.8. DIRECT SUM OF SUB SPACES

Definition : A vector space V is said to be the **direct sum** of its two subspaces W_1 and W_2 if every $v \in V$ can be **uniquely** expressed as

$$v = w_1 + w_2, \text{ where } w_1 \in W_1, w_2 \in W_2$$

We denote the direct sum of W_1 and W_2 by $W_1 \oplus W_2$ and write it as

$$V = W_1 \oplus W_2$$

The following example illustrate the difference between the linear sum and the direct sum of two sub-spaces.

I. Consider a vector space $V_3(\mathbf{R})$.

If $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$, $W_2 = \{(0, 0, c) : c \in \mathbf{R}\}$ are two subspaces of $V_3(\mathbf{R})$, then any vector $(a, b, c) \in V_3(\mathbf{R})$ can be written **uniquely** as:

$$(a, b, c) = (a, b, 0) + (0, 0, c)$$

$$\therefore V_3 = W_1 \oplus W_2$$

II. If we take $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$, $W_2 = \{(a, 0, c) : a, c \in \mathbf{R}\}$, which are also subspaces of $V_3(\mathbf{R})$, then any vector $(a, b, c) \in V_3(\mathbf{R})$ can be expressed as

$$(i) \quad (a, b, c) = \left(\frac{a}{2}, b, 0\right) + \left(\frac{a}{2}, 0, c\right), \text{ where } \left(\frac{a}{2}, b, 0\right) \in W_1 \text{ and } \left(\frac{a}{2}, 0, c\right) \in W_2; \text{ and also,}$$

$$(ii) \quad (a, b, c) = \left(\frac{a}{4}, b, 0\right) + \left(\frac{3a}{4}, 0, c\right), \text{ where } \left(\frac{a}{4}, b, 0\right) \in W_1 \text{ and } \left(\frac{3a}{4}, 0, c\right) \in W_2$$

(i) and (ii) show that elements of V_3 are **not uniquely** expressed as the sum of elements of W_1 and W_2 .

Thus in this case $V_3(\mathbf{R})$ is the **linear sum** of its two subspaces W_1 and W_2 but is not direct sum of two subspaces W_1 and W_2 .

1.9. DISJOINT SUBSPACES

Definition : Let $V(F)$ be a vector space. Two subspaces W_1 and W_2 of V are said to be disjoint if their intersection is the zero subspace. i.e., $W_1 \cap W_2 = \{0\}$.

1.10. THEOREM

The necessary and sufficient conditions for a vector space $V(F)$ to be a direct sum of its subspaces W_1 and W_2 are that

$$(i) \quad V = W_1 + W_2$$

$$(ii) \quad W_1 \cap W_2 = \{0\}$$

[M.D.U. 2008]

Proof. The conditions are necessary :

$$\text{Let} \quad V = W_1 \oplus W_2 \quad \dots(1)$$

$$\therefore \quad V = W_1 + W_2 \quad \dots(2) \text{ [By def. of direct sum]}$$

If possible, let $0 \neq v \in W_1 \cap W_2$

$$\Rightarrow \quad v \in W_1 \text{ and } v \in W_2$$

Also $v \in V$ and (2) gives

$$v = 0 + v, \text{ where } 0 \in W_1 \text{ and } v \in W_2$$

and

$$v = v + 0, \text{ where } v \in W_1 \text{ and } 0 \in W_2$$

which shows that $v \in V$ can be expressed in at least two different ways. This is contrary to (1).

Hence the only vector which is common to both W_1 and W_2 is 0 ,

$$\text{i.e.,} \quad W_1 \cap W_2 = \{0\}.$$

The conditions are sufficient :

$$\text{Let} \quad V = W_1 + W_2 \quad \dots(3)$$

$$\text{and} \quad W_1 \cap W_2 = \{0\} \quad \dots(4)$$

Now (3) implies that each element $v \in V$ can be expressed as sum of an element of W_1 and an element of W_2 .

We shall now show that this expression is unique. If possible, let

$$\therefore v = w_1 + w_2, w_1 \in W_1, w_2 \in W_2$$

and $v = w_1' + w_2', w_1' \in W_1, w_2' \in W_2$

$$\Rightarrow w_1 + w_2 = w_1' + w_2'$$

$$\Rightarrow w_1 - w_1' = w_2' - w_2$$

Since W_1 is a subspace,

$$\therefore w_1, w_1' \in W_1 \Rightarrow w_1 - w_1' \in W_1$$

Also $w_2, w_2' \in W_2 \Rightarrow w_2' - w_2 \in W_2$

$$\Rightarrow w_1 - w_1' = w_2' - w_2 \in W_1 \cap W_2$$

But $W_1 \cap W_2 = \{0\}$

Thus $w_1 - w_1' = 0 \Rightarrow w_1 = w_1'$

and $w_2' - w_2 = 0 \Rightarrow w_2 = w_2'$

which shows that expression for each $v \in V$ is unique and hence V is a direct sum of W_1 and W_2

i.e., $V = W_1 \oplus W_2$

SOLVED EXAMPLES

Example 1.

Let W_1, W_2, W_3 be the subspaces of $\mathbb{R}^3$ given by

$$W_1 = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}, W_2 = \{(x, y, z) \in \mathbb{R}^3 : x = z\}, W_3 = \{(0, 0, z) : z \in \mathbb{R}\}.$$

Prove that $\mathbb{R}^3 = W_1 + W_3$ and $\mathbb{R}^3 = W_2 + W_3$. Which of these sums is/are direct?

Solution. Each of $W_1 + W_3$ and $W_2 + W_3$ are subspaces of $\mathbb{R}^3$.

For any $(a, b, c) \in \mathbb{R}^3$,

$$(a, b, c) = (a, b, -a - b) + (0, 0, a + b + c) \in W_1 + W_3$$

Therefore $\mathbb{R}^3 = W_1 + W_3$

Also, $W_1 \cap W_3 = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0 \text{ and } x = y = 0\}$
 $= \{(0, 0, 0)\}.$

$$\therefore \mathbb{R}^3 = W_1 \oplus W_3$$

Similarly, for any $(a, b, c) \in \mathbb{R}^3$

$$(a, b, c) = (a, b, a) + (0, 0, c - a) \in W_2 + W_3$$

$$\therefore \mathbf{R}^3 = W_2 + W_3$$

Also,
$$W_2 \cap W_3 = \{(x, y, z) \in \mathbf{R}^3 : x = z \text{ and } x = y = 0\}$$

$$= \{(0, 0, 0)\}.$$

$$\therefore \mathbf{R}^3 = W_2 \oplus W_3$$

Hence, $W_1 + W_3$ and $W_2 + W_3$ are both direct sums.

Example 2.

Let V be the space of all functions from $\mathbf{R}$ to $\mathbf{R}$, and W_1 and W_2 be the subspace of V defined by $W_1 = \{f: f(x) = f(-x) \text{ for all } x \in \mathbf{R}\}$, $W_2 = \{f: f(-x) = -f(x) \text{ for all } x \in \mathbf{R}\}$. Show that $V = W_1 \oplus W_2$.

Solution. Suppose $f \in W_1 \cap W_2$. Then for all $x \in \mathbf{R}$

$$f(-x) = f(x) \text{ and } f(-x) = -f(x)$$

$$\Rightarrow f(x) = f(-x) = -f(x)$$

$$\Rightarrow f(x) = -f(x)$$

$$\Rightarrow f(x) = 0 \text{ for all } x \in \mathbf{R}$$

Thus f is a zero function and so $W_1 \cap W_2 = \{0\}$

Now, we claim that $V = W_1 + W_2$

Let $f \in V$. Define

$$g(x) = \frac{1}{2} \{f(x) + f(-x)\}$$

and

$$h(x) = \frac{1}{2} \{f(x) - f(-x)\}$$

Then (i) $f(x) = g(x) + h(x)$ for all $x \in \mathbf{R}$ i.e. $f = g + h$

$$(ii) \quad g(-x) = \frac{1}{2} \{f(-x) + f(x)\} = g(x) \Rightarrow g \in W_1$$

$$(iii) \quad h(-x) = \frac{1}{2} \{f(-x) - f(x)\} = -h(x) \Rightarrow h \in W_2$$

Thus for each $f \in V$, $f = g + h$ for some $g \in W_1$, $h \in W_2$.

Therefore $V = W_1 + W_2$ as $W_1 \cap W_2 = \{0\}$

Hence, $V = W_1 \oplus W_2$.

EXERCISE 1.3

1. Define linear sum of two subspaces and show that linear sum of two subspaces is also a subspace of the vector space $V(F)$.
2. In the vector space $V = \mathbf{R}^2(\mathbf{R})$; let $W_1 = \{(x, 0) : x \in \mathbf{R}\}$ and $W_2 = \{(0, x) : x \in \mathbf{R}\}$. Prove that W_1, W_2 are subspaces of $\mathbf{R}^2(\mathbf{R})$ and show that $V = W_1 \oplus W_2$.
3. Let $V = \mathbf{R}^2(\mathbf{R})$ be a vector space. Let $W_1 = \{(x, x) : x \in \mathbf{R}\}$, $W_2 = \{(x, 0) : x \in \mathbf{R}\}$. Prove that $V = W_1 \oplus W_2$.
4. Let V be the vector space of n -square matrices over a field $\mathbf{R}$. Let W_1 and W_2 be the subspaces of symmetric and skew-symmetric matrices of order n respectively. Show that V is the direct sum of W_1 and W_2 .

MISCELLANEOUS EXERCISE

1. Let $V = \{f(x) : f(x) \text{ is a polynomial of degree } \leq n \text{ (} n \text{ is a non-negative integer) or } f(x) \text{ is a zero polynomial}\}$. Show that V is a vector space under addition and scalar multiplication of polynomials.
2. Let $V = \{a + b\sqrt{2} + c\sqrt[3]{3} : a, b, c \in \mathbf{Q}\}$. Show that V is a vector space under addition and scalar multiplication of real numbers.
3. Let $V = \{f : f \text{ is a real valued continuous function defined on } [0, 1] \text{ and } f\left(\frac{2}{3}\right) = 2\}$. Show that V is not a vector space over $\mathbf{R}$ under addition and scalar multiplication defined as follows :

$$(f + g)(x) = f(x) + g(x) \text{ for all } f, g \in V$$

$$(af)(x) = af(x) \text{ for all } f \in V, a \in \mathbf{R}.$$

4. Which of the following sets form vector spaces over reals ? If not, why ?
 - (i) All n -rowed symmetric (skew-symmetric) matrices over $\mathbf{C}$.
 - (ii) All upper (lower) triangular matrices of order n over $\mathbf{R}$.
5. Show that $W = \{(a, b, c) : a, b, c \in \mathbf{Q}\}$ is not a subspace of $V_3(\mathbf{R})$.
6. Illustrate with the help of examples that if V is a set of all 2×2 matrices over $\mathbf{R}$, then
 - (i) the set of all 2×2 singular matrices
 - (ii) the set of all matrices A for which $A^2 = A$
 are not subspaces of V .

7. Let V be a vector space over $\mathbf{R}^3$. Examine whether the following are sub-spaces of V or not :
- (i) $W = \{(x, y, z) : x \in \mathbf{Z}\}$
 - (ii) $W = \{(x, y, z) : x \leq 0\}$
 - (iii) $W = \{(x, y, z) : x \geq y \geq z\}$
 - (iv) $W = \{(x, y, z) : x = y = z\}$
 - (v) $W = \{(x, y, z) : y + 4z = 0\}$
 - (vi) $W = \{(x, y, z) : x = y - z \text{ and } 2x + 3y - z = 0\}$
8. Examine whether the following are subspaces of $\mathbf{R}^4 = \{(x, y, z, w) : x, y, z, w \in \mathbf{R}\}$ or not:
- (i) $W = \{(m, m, m, m) : m \in \mathbf{R}\}$
 - (ii) $W = \{(m, n, m, n) : m, n \in \mathbf{Z}\}$
9. Examine whether the following set of vectors $u = (x_1, x_2, \dots, x_n) \in \mathbf{R}^n$ are subspaces of $\mathbf{R}^n (n \geq 3)$?
- (i) All u such that $x_1 \leq 0$
 - (ii) All u such that $x_1 x_2 = 0$
 - (iii) All u such that $x_3 \in \mathbf{Q}$
 - (iv) All u such that $x_2 = x_1^2$
 - (v) All u such that $x_1 + 2x_2 = 3x_3$
 - (vi) All u such that $x_1 + x_2 + \dots + x_n = \lambda$, a fixed real.
10. Let $V = \{f(x) : f(x) \text{ is a polynomial of degree } \leq 6\}$. Illustrate with the help of examples that which of the following are subspaces :
- (i) $W = \{f(x) : \deg f(x) = 5\}$
 - (ii) $W = \{f(x) : \deg f(x) \leq 3\}$
 - (iii) $W = \{f(x) : f(0) = 4\}$
 - (iv) $W = \{f(x) : f(2) = 0 = f(5)\}$
 - (v) $W = \{f(x) : \text{Coefficient of } x^2 \text{ is } 5 \text{ or } -5\}$
 - (vi) $W = \{f(x) : f(x) \text{ has -ve co-efficients}\}$

11. Let V be a vector space of all real valued continuous functions over $\mathbf{R}$. Show that the set W of solutions of differential equations $3 \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 2y = 0$ is a subspace of V .

ANSWERS

- | | | | |
|-----|----------|----------|--|
| 4. | (i) Yes | (ii) Yes | |
| 7. | (i) No | (ii) No | (iii) No |
| | (iv) Yes | (v) Yes | (vi) Yes |
| 8. | (i) Yes | (ii) No | |
| 9. | (i) No | (ii) No | (iii) No |
| | (iv) No | (v) Yes | (vi) Yes, if $\lambda = 0$; No, if $\lambda \neq 0$ |
| 10. | (i) No | (ii) Yes | (iii) No |
| | (iv) Yes | (v) No | (vi) No |